

* Recall:

(i) apparent magnitude: magnitude as we observe it from Earth

(ii) Absolute magnitude: magnitude if we placed the object at a distance 10 parsecs.

$$D = 10$$

(iii) Flux: $I_\lambda = \frac{L}{4\pi D^2}$ ← Luminosity
← Distance to the object



* Now, I introduced the empirical formula

$$\frac{I_2}{I_1} = (100)^{\frac{1}{5}(m_1 - m_2)}$$

Plug in $I_1 = \frac{L_1}{4\pi D_1^2}$, $I_2 = \frac{L_2}{4\pi D_2^2}$

$$\boxed{\frac{L_2 D_1^2}{L_1 D_2^2} = (100)^{\frac{1}{5}(m_1 - m_2)}}$$

* Why are we doing this? We want to find the relationship between the expression

given $\log_{10}\left(\frac{L_v}{(L_v)_\odot}\right) = \dots$

and the Distance to the object

(Object 1: Cepheid star $(L \rightarrow L_v)$)

Object 2: The Sun $\odot$ $L_2 \rightarrow L_{V\odot}$

→ We are given the absolute mag of the Sun

$$M_{V\odot} = 4.83$$

this means $D_2 = D_\odot = 10$ parsec

$$\frac{L_{V\odot}}{L_V} \frac{D^2}{10^2} = (100)^{\frac{1}{5}(V - M_{V\odot})}$$

Take $\log_{10}$:

$$\log_{10}\left(\frac{L_{V\odot}}{L_V}\right) + \log_{10}\left(\left(\frac{D}{10}\right)^2\right) = \frac{1}{5}(V - M_{V\odot}) \underbrace{\log_{10}(100)}_{=2}$$

→ Here I used

$$\log(A \cdot B) = \log(A) + \log(B)$$

→ This is almost what we want

* Recall:

$$\log\left(\frac{L_{V\odot}}{L_V}\right) = \log(L_{V\odot}) - \log(L_V)$$

multiply by (-1) on both sides

$$-\log\left(\frac{L_{V\odot}}{L_V}\right) = \log(L_V) - \log(L_{V\odot}) = \log\left(\frac{L_V}{L_{V\odot}}\right)$$

$$\text{thus } \log\left(\frac{L_{V\odot}}{L_V}\right) = -\log\left(\frac{L_V}{L_{V\odot}}\right)$$

thus, plugging this in

$$-\log\left(\frac{L_V}{L_{V\odot}}\right) + 2\log\left(\frac{D}{10}\right) = \frac{2}{5}(V - M_{V\odot})$$

→ Plug in the given expression

$$-\left[0.9\log(P) + 2.75\right] + 2\log\left(\frac{D}{10}\right) = \frac{2}{5}(V - M_{V\odot})$$

$$2 \log\left(\frac{D}{10}\right) = 0.9 \log(P) + 2.75 + \frac{2}{5}(V - M_{V0})$$

- Can now solve for D and plug in values for V , M_{V0} and P .
→ Make sure you convert $P = 224$ hr into days!

